

E20471250_1

The equation $x^4 + x + y^3 + 4y + 1 = 0$ has no integer solutions

Size $h = 35$, length $l = 11$

Theorem 1. *The equation*

$$x^4 + x + y^3 + 4y + 1 = 0 \tag{1}$$

has no integer solutions.

The proof uses four polynomial identities of the form $H_k T_k = U_k^2 + d_k V_k^2$ on the curve (1), where $d_1 = 2H_2$, $d_2 = -2H_1$, $d_3 = H_4$, and $d_4 = H_3$, and the product formula for two Hilbert symbols. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Legendre symbol and of the Hilbert symbol.

1 Legendre symbols and Hilbert symbols

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. It is multiplicative in a , and it depends only on a modulo p [1, Chapter 5].

For a prime p , let $\mathbb{Q}_p$ be the field of p -adic numbers, and put $\mathbb{Q}_\infty = \mathbb{R}$. For v a prime or $v = \infty$, and non-zero $a, b \in \mathbb{Q}$, the Hilbert symbol $(a, b)_v$ is 1 if the equation $z^2 = ar^2 + bs^2$ has a solution $(r, s, z) \neq (0, 0, 0)$ in $\mathbb{Q}_v$, and -1 otherwise. In particular, $(a, b)_v = 1$ for every v if this equation has a non-zero solution in $\mathbb{Q}$. For an odd integer t , we put $\varepsilon(t) = (t - 1)/2$ and $\omega(t) = (t^2 - 1)/8$. We use the following facts [2, Chapter III, Proposition 2 and Theorems 1 and 3].

Theorem 2. *Let a, a' and b be non-zero integers, and let v be a prime or $v = \infty$.*

- (i) $(a, b)_\infty = -1$ if and only if $a < 0$ and $b < 0$.
- (ii) Let p be an odd prime, and write $a = p^\alpha u$ and $b = p^\beta w$ with $p \nmid uw$. Then $(a, b)_p = (-1)^{\alpha\beta\varepsilon(p)}(u/p)^\beta(w/p)^\alpha$.
- (iii) Write $a = 2^\alpha u$ and $b = 2^\beta w$ with u and w odd. Then $(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(w) + \alpha\omega(w) + \beta\omega(u)}$.
- (iv) $(a, b)_v = 1$ for all but finitely many v , and $\prod_v (a, b)_v = 1$, the product over all primes v and $v = \infty$.
- (v) $(a, b)_v = (b, a)_v$ and $(aa', b)_v = (a, b)_v(a', b)_v$.

Lemma 3. *Let p be an odd prime, and let a, b, r, c, t be integers with $a \neq 0$, $b \neq 0$ and $at = r^2 - bc^2$. If $p \nmid b$, and p does not divide both r and c , then $(a, b)_p = 1$.*

Proof. If $p \nmid a$, this is Theorem 2(ii) with $\alpha = \beta = 0$. If $p \mid a$, then $r^2 \equiv bc^2 \pmod{p}$. If $p \mid c$, then $p \mid r$, which is excluded. Hence $p \nmid c$, so $b \equiv (rc')^2 \pmod{p}$, where $cc' \equiv 1 \pmod{p}$, and $(b/p) = 1$ as $p \nmid b$. Theorem 2(ii) with $\beta = 0$ gives $(a, b)_p = (b/p)^{v_p(a)} = 1$. $\square$

Lemma 4. *Let a, b, r, c, t be integers with $at = r^2 - bc^2 \neq 0$ and $b \neq 0$. Then $(a, b)_v = (t, b)_v$ for every v .*

Proof. The triple $(r, s, z) = (1, c, r)$ is a non-zero rational solution of $z^2 = (at)r^2 + bs^2$, so $(at, b)_v = 1$. By Theorem 2(v), $(a, b)_v(t, b)_v = (at, b)_v = 1$, and both symbols are ± 1 . $\square$

Lemma 5. *Let a and b be non-zero integers. If p is an odd prime with $p \nmid a$ and $(a/p) = 1$, then $(a, b)_p = 1$. If $a \equiv 1 \pmod{8}$, then $(a, b)_2 = 1$.*

Proof. The first claim is Theorem 2(ii) with $\alpha = 0$. For the second, $\varepsilon(a)$ and $\omega(a)$ are even, and Theorem 2(iii) with $\alpha = 0$ gives $(a, b)_2 = 1$. $\square$

2 The auxiliary polynomials

Put $F(x, y) = x^4 + x + y^3 + 4y + 1$, so that (1) is the equation $F(x, y) = 0$. For the first identity, we define

$$\begin{aligned} H_1 &= -y, \\ T_1 &= 4y^2 + 18, \\ U_1 &= 2x^2, \\ V_1 &= 1, \\ G_1 &= -4, \end{aligned}$$

for the second identity, we define

$$\begin{aligned} H_2 &= 2x - y + 2, \\ T_2 &= 16x^2 + 8xy + 32x + 4y^2 + 8y + 30, \\ U_2 &= 2x^2 + 8x + 8, \\ V_2 &= 1, \\ G_2 &= -4, \end{aligned}$$

for the third identity, we define

$$\begin{aligned} H_3 &= -x - 2y, \\ T_3 &= 6x - 6y + 10, \\ U_3 &= -3y, \\ V_3 &= 1, \\ G_3 &= 0, \end{aligned}$$

for the fourth identity, we define

$$\begin{aligned} H_4 &= -6x^2 - 6xy - 10x + 3y^2 - 20y, \\ T_4 &= -6x - 3y + 7, \\ U_4 &= -3x^2 - 6x - 9y + 3, \\ V_4 &= 5, \\ G_4 &= -9, \end{aligned}$$

for the auxiliary polynomials H_1 and H_2 together, we define

$$\lambda_1 = x^3 - x^2 + x + 2y^2 + 8,$$

$$\begin{aligned}\mu_1 &= -x^3 + x^2 - x, \\ \nu_1 &= 2,\end{aligned}$$

for the auxiliary polynomials H_3 and H_4 together, we define

$$\begin{aligned}\lambda_2 &= -30x^4 + 12x^3y - 104x^3 + 6x^2y - 64x^2 + 9xy^2 - 6xy + 76x + 9y^2 - 12y \\ &\quad + 116, \\ \mu_2 &= 8x^3 + 7x^2 - x - 8, \\ \nu_2 &= 18x + 18,\end{aligned}$$

and, for the signs, we also use

$$\begin{aligned}Q_1 &= y^2 + 4, \\ M_1 &= x^4 + x + 1, \\ E_1 &= 16, \\ Q_2 &= 4x^2 + 2xy + 8x + y^2 + 2y + 8, \\ M_2 &= x^4 + 8x^3 + 24x^2 + 33x + 17, \\ E_2 &= 12x^2 + 24x + 28, \\ Q_3 &= x^2 - 2xy + 4y^2 + 16, \\ M_3 &= 8x^4 - x^3 - 8x + 8, \\ E_3 &= 12x^2 + 256, \\ Q_5 &= 36x^2 - 18xy - 84x + 9y^2 + 21y + 85, \\ M_5 &= 27x^4 - 216x^3 + 756x^2 - 1071x + 622, \\ E_5 &= 972x^2 - 2268x + 2619.\end{aligned}$$

Expanding both sides, one checks the identities

$$H_1T_1 - U_1^2 - 2H_2V_1^2 = G_1F, \quad (2)$$

$$H_2T_2 - U_2^2 + 2H_1V_2^2 = G_2F, \quad (3)$$

$$H_3T_3 - U_3^2 - H_4V_3^2 = G_3F, \quad (4)$$

$$H_4T_4 - U_4^2 - H_3V_4^2 = G_4F, \quad (5)$$

$$\lambda_1H_1 + \mu_1H_2 + \nu_1F = 2, \quad (6)$$

$$\lambda_2H_3 + \mu_2H_4 + \nu_2F = 2 \cdot 3^2, \quad (7)$$

$$-F = Q_1H_1 - M_1, \quad (8)$$

$$4Q_1 = (2y)^2 + E_1. \quad (9)$$

$$-F = Q_2H_2 - M_2, \quad (10)$$

$$4Q_2 = (2x + 2y + 2)^2 + E_2. \quad (11)$$

$$-8F = Q_3H_3 - M_3, \quad (12)$$

$$16Q_3 = (-2x + 8y)^2 + E_3. \quad (13)$$

$$-27F = Q_5(-T_{B_2}) - M_5, \quad (14)$$

$$36Q_5 = (-18x + 18y + 21)^2 + E_5. \quad (15)$$

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) . Put

$$\begin{aligned} A_1 &= -d_2 = 2H_1, & B_1 &= -d_1 = -2H_2, & T_{A_1} &= 2T_1, & T_{B_1} &= -2T_2, \\ A_2 &= -d_4 = -H_3, & B_2 &= -d_3 = -H_4, & T_{A_2} &= -T_3, & T_{B_2} &= -T_4. \end{aligned}$$

By (2)–(5), after multiplying by squares of the constants c in $d_k = cH_{k'}$ if necessary, we obtain

$$A_1 T_{A_1} = (2U_1)^2 - B_1 (2V_1)^2, \quad B_1 T_{B_1} = (2U_2)^2 - A_1 (2V_2)^2. \quad (16)$$

$$A_2 T_{A_2} = U_3^2 - B_2 V_3^2, \quad B_2 T_{B_2} = U_4^2 - A_2 V_4^2. \quad (17)$$

Step 1: signs. *The sign of H_1 .* Since $F(x, y) = 0$, (8) gives $Q_1 H_1 = M_1$, and M_1 and E_1 depend only on x . We show that $M_1 > 0$ and $E_1 > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_1 and E_1 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_1 > 0$ and $E_1 > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_1 > 0$ and $E_1 > 0$. Hence $M_1 > 0$ and $E_1 > 0$, so (9) gives $4Q_1 \geq E_1 > 0$, and $Q_1 H_1 = M_1 > 0$ gives $H_1 > 0$.

The sign of H_2 . Since $F(x, y) = 0$, (10) gives $Q_2 H_2 = M_2$, and M_2 and E_2 depend only on x . We show that $M_2 > 0$ and $E_2 > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -4$, write $x = -4 - z$ with $z \geq 0$. In both cases, expanding M_2 and E_2 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_2 > 0$ and $E_2 > 0$. It remains to consider $-3 \leq x \leq 0$. For $x = -1$ and $x = 0$ we have $M_2 > 0$ and $E_2 > 0$. For $x = -2$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , as one checks for the 7 residues of y modulo 7, so (1) has no solution with this x . For $x = -3$, we have $F(x, y) \not\equiv 0 \pmod{13}$ for every y , as one checks for the 13 residues of y modulo 13, so (1) has no solution with this x . Hence $M_2 > 0$ and $E_2 > 0$, so (11) gives $4Q_2 \geq E_2 > 0$, and $Q_2 H_2 = M_2 > 0$ gives $H_2 > 0$.

The sign of H_3 . Since $F(x, y) = 0$, (12) gives $Q_3 H_3 = M_3$, and M_3 and E_3 depend only on x . We show that $M_3 > 0$ and $E_3 > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_3 and E_3 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_3 > 0$ and $E_3 > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_3 > 0$ and $E_3 > 0$. Hence $M_3 > 0$ and $E_3 > 0$, so (13) gives $16Q_3 \geq E_3 > 0$, and $Q_3 H_3 = M_3 > 0$ gives $H_3 > 0$.

By these signs, $A_1 > 0$, $B_1 < 0$, and $A_2 < 0$.

The sign of B_2 . Since $F(x, y) = 0$, (14) gives $Q_5(-T_{B_2}) = M_5$, and M_5 and E_5 depend only on x . We show that $M_5 > 0$ and $E_5 > 0$. If $x \geq 2$, write $x = 2 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_5 and E_5 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_5 > 0$ and $E_5 > 0$. It remains to consider $0 \leq x \leq 1$. For $x = 0$ and $x = 1$ we have $M_5 > 0$ and $E_5 > 0$. Hence $M_5 > 0$ and $E_5 > 0$, so (15) gives $36Q_5 \geq E_5 > 0$, and $Q_5(-T_{B_2}) = M_5 > 0$ gives $(-T_{B_2}) > 0$. As $A_2 < 0$ and $V_4 \neq 0$, (17) gives $B_2 T_{B_2} = U_4^2 + |A_2| V_4^2 > 0$. Hence $B_2 < 0$.

Hence Theorem 2(i) gives $(A_1, B_1)_\infty = 1$ (as $A_1 > 0$) and $(A_2, B_2)_\infty = -1$ (as $A_2 < 0$ and $B_2 < 0$).

Step 2: primes outside S . Let $S = \{2, 3, 5\}$ be the set of primes dividing 2, the constants 1 and 2 in the $d_k = cH_{k'}$, $V_1 = 1$, $V_2 = 1$, $V_3 = 1$, and $V_4 = 5$, and the right-hand sides of (6) and (7). Let $p \notin S$ be an odd prime and $j \in \{1, 2\}$; we show that $(A_j, B_j)_p = 1$. If $p \nmid A_j B_j$, this follows from Theorem 2(ii) with $\alpha = \beta = 0$. For $j = 1$: If $p \mid A_1$, then $p \mid H_1$, so $p \nmid H_2$ by (6) and $p \nmid B_1$; as $p \nmid 2V_1$, Lemma 3 with $(a, b, r, c, t) = (A_1, B_1, 2U_1, 2V_1, T_{A_1})$ and (16) gives $(A_1, B_1)_p = 1$; if $p \mid B_1$, then $p \mid H_2$, so $p \nmid H_1$ by (6) and $p \nmid A_1$; as $p \nmid 2V_2$, Lemma 3 with $(a, b, r, c, t) = (B_1, A_1, 2U_2, 2V_2, T_{B_1})$ and (16) gives $(B_1, A_1)_p = 1$, which equals $(A_1, B_1)_p$ by

Theorem 2(v). For $j = 2$: If $p \mid A_2$, then $p \mid H_3$, so $p \nmid H_4$ by (7) and $p \nmid B_2$; as $p \nmid V_3$, Lemma 3 with $(a, b, r, c, t) = (A_2, B_2, U_3, V_3, T_{A_2})$ and (17) gives $(A_2, B_2)_p = 1$; if $p \mid B_2$, then $p \mid H_4$, so $p \nmid H_3$ by (7) and $p \nmid A_2$; as $p \nmid V_4$, Lemma 3 with $(a, b, r, c, t) = (B_2, A_2, U_4, V_4, T_{B_2})$ and (17) gives $(B_2, A_2)_p = 1$, which equals $(A_2, B_2)_p$ by Theorem 2(v).

Step 3: the primes in S . By Step 1, the A_j and B_j are non-zero. If $T_{A_j} \neq 0$, then (16)–(17) and Lemma 4 give $(A_j, B_j)_v = (T_{A_j}, B_j)_v$, and if $T_{B_j} \neq 0$, they give $(A_j, B_j)_v = (B_j, A_j)_v = (T_{B_j}, A_j)_v = (A_j, T_{B_j})_v$, using Theorem 2(v). For a non-zero integer X and a prime p , we write $X \in p^{2i}(\pm)$ or $X \in p^{2i+1}(\pm)$, when p is odd, if $X = p^j u$ with $p \nmid u$, $(u/p) = \pm 1$, and j even or odd, respectively; and we write $X \in 2^{2i} \cdot u$ or $X \in 2^{2i+1} \cdot u$ if $X = 2^j u'$ with $u' \equiv u \pmod{8}$ and j even or odd. If $X \not\equiv 0 \pmod{p^e}$ and $v_p(X) \leq e - 1$, or $v_p(X) \leq e - 3$ if $p = 2$, then this class is determined by X modulo p^e . By Theorem 2(ii) and (iii), the classes of X and Y determine $(X, Y)_p$.

The prime 2. Checking the 16^2 residue pairs modulo 16, we find that (1) has exactly 16 solutions modulo 16, and that at each of them, for each j , one of the cases listed for $p = 2$ and j in Table 1 holds; moreover, the combination of these cases is one of the combinations listed there. Each case gives the stated value of $(A_j, B_j)_2$. Hence $\sigma_2 \in V_2$, with V_2 as in Step 4.

The prime 3. Checking the 3^2 residue pairs modulo 3, we find that (1) has exactly 3 solutions modulo 3, and that at each of them, for each j , one of the cases listed for $p = 3$ and j in Table 1 holds. Each case gives the stated value of $(A_j, B_j)_3$. Hence $\sigma_3 \in V_3$, with V_3 as in Step 4.

The prime 5. Checking the 5^2 residue pairs modulo 5, we find that (1) has exactly 6 solutions modulo 5, and that at each of them, for each j , one of the cases listed for $p = 5$ and j in Table 1 holds. Each case gives the stated value of $(A_j, B_j)_5$. Hence $\sigma_5 \in V_5$, with V_5 as in Step 4.

Step 4: contradiction. For $p \in S$ and $p = \infty$, put

$$\sigma_p = ((A_1, B_1)_p, (A_2, B_2)_p).$$

By Theorem 2(iv), applied to each j , and by Step 2, the componentwise product of the σ_p with $p \in S$ is σ_∞ , which is $(1, -1)$ by Step 1. By Step 3, $\sigma_p \in V_p$, where

$$\begin{aligned} V_2 &= \{(1, 1), (-1, 1), (-1, -1)\}, \\ V_3 &= \{(1, 1)\}, \\ V_5 &= \{(1, 1)\}. \end{aligned}$$

The componentwise products of one vector from each of V_3 and V_5 are $(1, 1)$, so σ_2 would have to be $(1, -1)$, but this vector is not in V_2 . This contradiction proves the theorem. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.
- [2] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, New York, 1973.

Table 1: The cases in Step 3. In each case, the symbol $(A_j, B_j)_p$ is computed as the stated symbol, whose entries lie in the stated classes; an entry “any” means that the other entry is a unit square, so that Lemma 5 applies.

	symbol	first entry	second entry	value
$p = 2$, solutions modulo 16				
$j = 1$				
1	(A_1, B_1)	$2^{2i+1} \cdot 1$	$2^{2i+1} \cdot 5$	-1
2	(A_1, B_1)	$2^{2i+1} \cdot 1$	$2^{2i+1} \cdot 7$	1
3	(A_1, B_1)	$2^{2i+1} \cdot 3$	$2^{2i+1} \cdot 1$	-1
4	(A_1, B_1)	$2^{2i+1} \cdot 3$	$2^{2i+1} \cdot 7$	1
5	(A_1, B_1)	$2^{2i+1} \cdot 5$	$2^{2i+1} \cdot 1$	-1
6	(A_1, B_1)	$2^{2i+1} \cdot 5$	$2^{2i+1} \cdot 3$	1
7	(A_1, B_1)	$2^{2i+1} \cdot 7$	$2^{2i+1} \cdot 3$	1
8	(A_1, B_1)	$2^{2i+1} \cdot 7$	$2^{2i+1} \cdot 5$	-1
$j = 2$				
1	(A_2, B_2)	any	$2^{2i} \cdot 1$	1
2	(A_2, B_2)	$2^{2i} \cdot 1$	$2^{2i} \cdot 3$	1
3	(A_2, B_2)	$2^{2i} \cdot 3$	$2^{2i} \cdot 7$	-1
4	(A_2, B_2)	$2^{2i} \cdot 5$	$2^{2i} \cdot 3$	1
5	(A_2, B_2)	$2^{2i} \cdot 7$	$2^{2i} \cdot 7$	-1
6	(A_2, B_2)	$2^{2i+1} \cdot 1$	$2^{2i} \cdot 1$	1
7	(A_2, B_2)	$2^{2i+1} \cdot 3$	$2^{2i} \cdot 1$	1
8	(A_2, B_2)	$2^{2i+1} \cdot 5$	$2^{2i} \cdot 1$	1
9	(A_2, B_2)	$2^{2i+1} \cdot 7$	$2^{2i} \cdot 1$	1
combinations of cases ($j = 1, j = 2$) and the vector σ_2				
	$(1, 6) \mapsto (-1, 1)$	$(1, 8) \mapsto (-1, 1)$		
	$(2, 2) \mapsto (1, 1)$	$(3, 5) \mapsto (-1, -1)$		
	$(4, 1) \mapsto (1, 1)$	$(5, 7) \mapsto (-1, 1)$		
	$(5, 9) \mapsto (-1, 1)$	$(6, 4) \mapsto (1, 1)$		
	$(7, 1) \mapsto (1, 1)$	$(8, 3) \mapsto (-1, -1)$		
$p = 3$, solutions modulo 3				
$j = 1$				
1	$3 \nmid A_1 B_1$			1
2	(A_1, B_1)	any	$3^{2i}(+)$	1
$j = 2$				
1	$3 \nmid A_2 B_2$			1
$p = 5$, solutions modulo 5				
$j = 1$				
1	$5 \nmid A_1 B_1$			1
2	(T_{A_1}, B_1)	$5^{2i}(+)$	$5^{2i}(+)$	1
$j = 2$				
1	$5 \nmid A_2 B_2$			1
2	(T_{A_2}, B_2)	$5^{2i}(-)$	$5^{2i}(+)$	1
3	(T_{A_2}, B_2)	$5^{2i}(+)$	$5^{2i}(+)$	1